

UPPER EXTREMITY FRACTURES — PART 2 (Humeral Shaft → Hand)

Humeral shaft

- Common diaphyseal fracture; **radial nerve injury risk**.
- **Holstein-Lewis fracture**: spiral fracture of distal third → **radial nerve neuropraxia** (wrist drop + MCP extensor paralysis).
- **Operative indications**: multiple injuries, open, segmental, intra-articular extension, pathological, **floating elbow** (humeral + forearm fracture), **radial nerve palsy after manipulation**, non-union, nursing problems.

Radius & ulna (both-bone forearm)

- **Adults**: nearly all → **ORIF** (non-op rare).
- **Children**: mostly non-operative if adequate reduction; greenstick; bayonet ok if <10yrs. Operative if angulation >15°/rotation >45° (<10yrs), >10°/30° (>10yrs), bayonet >10yrs, open, both-bone >13yrs.

Fracture-dislocations

- **Monteggia**: **proximal 1/3 ulna fracture + radial head dislocation**. Adults → ORIF ulna; children → closed reduction if radiocapitellar stable.
- **Galeazzi**: **distal 1/3 radius fracture + DRUJ injury**. Adults → ORIF radius + assess DRUJ; children → closed reduction mostly.

TFCC injuries

- **Ulnar-sided wrist pain**, worse on ulnar deviation, **positive fovea sign**; MRI confirms. Type 1 traumatic (fall on extended wrist, pronation), Type 2 degenerative (positive ulnar variance, ulnocarpal impaction). Tx: conservative (NSAIDs, immobilize) → debridement/repair/ulnar shortening.

Distal radius (adults)

- **Most common orthopaedic injury**; fall on outstretched hand.
- **Non-operative**: extra-articular, <5mm shortening, dorsal angulation <5°.
- **ORIF indications**: instability, dorsal angulation >5°, comminution, **intra-articular displacement >2mm**, shortening >5mm, **Barton's (volar/dorsal)**, **Smith's (volar)**, die-punch, **critical corner (volar ulnar)** fragment. Pre-reduction X-ray = best stability predictor.

Distal forearm (children)

- Commonest paediatric forearm site; fall on outstretched hand. Mostly closed reduction + cast. Torus/buckle → short arm cast. **CRPP** if unstable or NV compromise; **ORIF** for displaced Salter-Harris III/IV.

Carpal injuries

- Common: sprains, **scaphoid fracture**, TFCC/DRUJ, lunate/perilunate dislocation, carpal collapse.

Scaphoid fracture

- **75% of carpal fractures**; fall on outstretched hand. Locations: **waist 65%, proximal 25%, distal 10%**.
- **Blood supply: dorsal carpal branch (radial artery) enters distally → supplies proximal 80% via RETROGRADE flow → proximal pole AVN/non-union risk** (watershed).
- Non-op: stable non-displaced (majority); **suspected with normal X-ray → thumb spica + re-image 12–21 days**. Operative if displaced, humpback >15°, perilunate, comminuted.

Lunate/perilunate dislocation

- High-energy, dorsiflexion; ligament tears. **Perilunate** = carpus displaced dorsally (lunate stays with radius); **lunate dislocation** = lunate displaced anteriorly; **trans-scaphoid perilunate** = scaphoid waist fracture. Urgent reduction.

Metacarpal fractures

- Shaft (transverse/spiral — rotation risk → fix). **Neck = "boxer's fracture" (5th MC)** — volar angulation. Base usually stable.

Thumb metacarpal

- **Bennett's**: oblique intra-articular base of 1st MC, **unstable**, extends into CMC joint (punching). Reduce + plaster/fixation.
- **Rolando's**: comminuted T/Y intra-articular base of 1st MC; K-wire/plate/ext fix.

Gamekeeper's / skier's thumb

- **Ulnar collateral ligament of thumb MCP**; fall onto extended thumb (hyperabduction). Partial (stable in extension) vs complete (unstable all positions). **Stener lesion** = ligament trapped in front of adductor aponeurosis → won't heal → needs repair.

High-yield one-liners

- **Holstein-Lewis** = distal third spiral humeral fracture + radial nerve palsy.
- **Monteggia** = proximal ulna + radial head dislocation; **Galeazzi** = distal radius + DRUJ.
- **Scaphoid**: waist commonest, retrograde blood supply → proximal pole AVN; suspect with normal X-ray → thumb spica + re-image.
- **Distal radius** = commonest injury; **Barton's** (intra-articular), **Smith's** (volar), **boxer's** = 5th MC neck.
- **Bennett's** (unstable) vs **Rolando's** (comminuted) thumb base; **Gamekeeper's** = UCL + Stener lesion.

UPPER EXTREMITY FRACTURES PART 2 — Revision Table

Bold = high-yield. Humeral shaft → hand. Dr Zaid Al-Bachachy.

SHAFT & FOREARM

Injury	Key points
Humeral shaft	Radial nerve risk; Holstein-Lewis = distal third spiral + radial nerve palsy (wrist drop) ; surgery if floating elbow, open, post-manipulation palsy
Both-bone forearm	Adults → ORIF nearly all ; children mostly non-op (greenstick)
Monteggia	Proximal 1/3 ulna + radial head dislocation → ORIF (adult)
Galeazzi	Distal 1/3 radius + DRUJ injury → ORIF radius + assess DRUJ
TFCC	Ulnar wrist pain, fovea sign ; MRI; conservative → debridement/repair

DISTAL RADIUS & CARPUS

Injury	Key points
Distal radius (adult)	Most common ortho injury ; ORIF if instability, dorsal angulation >5°, intra-articular >2mm, Barton's/Smith's , critical corner
Distal forearm (child)	Closed reduction + cast; CRPP if unstable; ORIF SH III/IV
Scaphoid	75% carpal fractures; waist 65%; retrograde supply → proximal pole AVN/non-union ; suspect with normal X-ray → thumb spica + re-image 12–21d
Lunate/perilunate	High-energy dorsiflexion; trans-scaphoid perilunate ; urgent reduction

HAND

Injury	Key points
Metacarpal	Shaft (rotation → fix); neck = boxer's (5th MC) ; base stable
Bennett's	Oblique intra-articular base 1st MC, unstable , into CMC (punching)
Rolando's	Comminuted T/Y base 1st MC
Gamekeeper's/skier's thumb	UCL of thumb MCP; Stener lesion (trapped by adductor aponeurosis → won't heal → repair)